

Technology Stack

Date	1 July 2026
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	Flask (Jinja2 Templates), HTML/CSS	Server-rendered pages for the Home, Prediction Input, and Flood/No-Flood Result screens keep the app simple and fast to deploy.
2	Backend / Server-Side	Python, Flask	Lightweight framework that integrates directly with Python's ML libraries for fast processing and simple routing to the prediction model.
3	Machine Learning	Scikit-learn, XGBoost, NumPy, Pandas, Matplotlib, Seaborn	Powers the Decision Tree, Random Forest, KNN, and XGBoost models, plus the data analysis, cleaning, and visualization steps.
4	Model Storage	Joblib / Pickle (floods.save)	Saves the trained XGBoost model and scaler so predictions can be served instantly without retraining.
5	Database / Data Storage	Kaggle Flood Prediction Dataset (CSV)	Open-source rainfall and weather dataset used to train and validate the models; no live database is required for predictions.
6	Cloud / Hosting / Deployment	IBM Cloud	Enables scalable, cost-effective deployment so disaster management teams can access flood risk predictions from any location.
7	Version Control & CI/CD	GitHub	Supports seamless collaboration and change tracking among team members throughout development.
8	Third-Party APIs / Other	Anaconda Navigator	Simplifies setup and management of the

	Tools		Python environment and required libraries across the team.
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